

Equations of a Liquid State Machine

A spiking reservoir computer, end to end

Overview

A liquid state machine (LSM) is a reservoir computer whose reservoir—the *liquid*—is a recurrent network of spiking neurons. The recurrent weights W and the input weights W_{in} are fixed (random); only a linear readout is trained. The pipeline below carries an input $u(t)$ through the membrane dynamics of each neuron, into spikes, through a synaptic filter that turns those spikes into a continuous *trace*, and finally into the output. The trace $r(t)$ does double duty: it is both the recurrent drive inside the liquid and the state vector read by the output.

Notation

Symbol	Meaning
$V_i(t)$	membrane potential of neuron i
$\tau_m = RC$	<i>membrane</i> time constant (leak of V_i)
$V_{rest}, V_{th}, V_{reset}$	rest, threshold, and reset potentials
τ_{ref}	refractory period
$s_i(t)$	spike train of neuron i (sum of impulses)
$t_i^{(k)}$	time of the k -th spike of neuron i
$\kappa(s)$	synaptic kernel; $s \geq 0$ is time elapsed since a spike
$\tau_s (\tau_r, \tau_d)$	<i>synaptic</i> time constant(s); $\neq \tau_m$
$r_i(t)$	synaptic trace of neuron i (filtered spike train)
W	recurrent (reservoir) weight matrix, fixed
W_{in}	input weight matrix, fixed
$u(t)$	external input
W_{out}	readout weight matrix, trained
$\hat{y}(t)$	network output

1 Membrane potential (LIF), per neuron

Between spikes, each neuron i is a leaky integrator driven by its total input current $I_i(t)$:

$$\tau_m \frac{dV_i}{dt} = -(V_i(t) - V_{rest}) + R I_i(t). \quad (1)$$

2 Spike generation (threshold and reset)

When the membrane reaches threshold, the neuron emits a spike and resets:

$$V_i(t) \geq V_{th} \implies \text{record } t_i^{(k)}, \quad V_i \rightarrow V_{reset}, \quad \text{hold for } \tau_{ref}. \quad (2)$$

3 Spike train

The spike train is a sum of time-shifted impulses (Dirac deltas) at the firing times:

$$s_i(t) = \sum_k \delta(t - t_i^{(k)}). \quad (3)$$

4 Synaptic trace (filtered spike train)

Each spike is convolved with a causal synaptic kernel κ , replacing every impulse with a decaying bump and summing them:

$$r_i(t) = (\kappa * s_i)(t) = \sum_k \kappa(t - t_i^{(k)}). \quad (4)$$

Common choices of kernel (all causal: $\kappa(s) = 0$ for $s < 0$):

$$\text{single exponential: } \kappa(s) = \frac{1}{\tau_s} e^{-s/\tau_s} \iff \tau_s \dot{r}_i = -r_i + s_i(t), \quad (5)$$

$$\text{double exponential: } \kappa(s) \propto e^{-s/\tau_d} - e^{-s/\tau_r} \quad (\tau_r < \tau_d), \quad (6)$$

$$\text{alpha function: } \kappa(s) = \frac{s}{\tau_s} e^{1-s/\tau_s}. \quad (7)$$

Here s is the time *elapsed since a spike*, and τ_s (or τ_d) is the *synaptic* decay constant—distinct from the membrane τ_m .

5 Input current: recurrent + external

The current driving neuron i is the weighted sum of its presynaptic neighbours' traces, plus the external drive (and an optional tonic background):

$$I_i(t) = \underbrace{\sum_j W_{ij} r_j(t)}_{\text{recurrent}} + \underbrace{\sum_m (W_{\text{in}})_{im} u_m(t)}_{\text{external}} + I_{\text{tonic}}. \quad (8)$$

A presynaptic spike from neuron j becomes a current bump $r_j(t)$ in neuron i , scaled by W_{ij} ; the trace is therefore the physical channel by which neurons communicate, not merely a readout convenience.

Conductance-based variant. In conductance-based models (e.g. the multiplexing-oscillations network), the recurrent term carries a voltage-dependent driving force:

$$I_i^{\text{rec}}(t) = \sum_j g_{ij} r_j(t) (E_j - V_i(t)), \quad (9)$$

where E_j is the synapse's reversal potential ($E \approx 0$ for excitation, $E \approx -80$ mV for inhibition). The weighted-filtered-spike structure is identical.

6 Readout

The output is a linear map on the population trace vector $r(t) = (r_1(t), \dots, r_N(t))^\top$:

$$\hat{y}(t) = W_{\text{out}}^\top r(t). \quad (10)$$

7 Training (readout only)

The liquid (W, W_{in}) is frozen; only W_{out} is learned, e.g. by online recursive least squares (RLS):

$$e(t) = \hat{y}(t) - y(t), \quad (11)$$

$$W_{\text{out}} \leftarrow W_{\text{out}} - P(t) r(t) e(t)^\top, \quad (12)$$

$$P(t) = P(t - \Delta t) - \frac{P r r^\top P}{1 + r^\top P r}, \quad P(0) = \frac{1}{\alpha} I. \quad (13)$$

Freezing the liquid is what makes the simple online rule well posed: the echo-state / separation property guarantees that $r(t)$ is a stable function of the input history, so fitting W_{out} is a stationary regression.

Remarks

1. **Two timescales, not one.** $\tau_m = RC$ governs the membrane (how fast V_i forgets its input); τ_s governs the synapse (how fast a trace r_i decays). They are independent parameters with different physiological meanings.
2. **Three notions of “state.”** $V_i(t)$ is the hidden internal state (it has resets and is never read out); $s_i(t)$ is the spikes; $r_i(t)$ is the usable continuous state that both drives the recurrence (Eq. 8) and feeds the readout (Eq. 10).
3. **Relation to the rate ESN.** Equations 1–5 collapse into the single nonlinearity $F(\cdot)$ of the echo-state-network map $x(t+1) = F(Wx(t) + W_{\text{in}}u(t))$ once the spike-and-filter mechanism is replaced by a frequency–current (f–I) gain curve; the recurrent term $\sum_j W_{ij}r_j(t)$ is exactly the $Wx(t)$ that survives that reduction.